

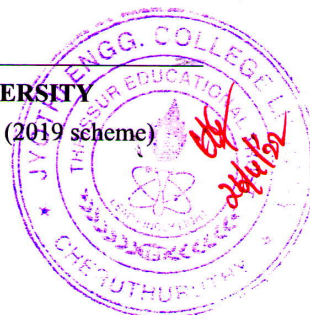
Reg No.:

Name:

1100EET307122103

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)



Course Code: EET307

Course Name: SYNCHRONOUS AND INDUCTION MACHINES

(GRAPH SHEET NEEDED)

Max. Marks: 100

Duration: 3 Hours

**PART A**

(Answer all questions; each question carries 3 marks)

- |    |                                                                                                             | Marks |
|----|-------------------------------------------------------------------------------------------------------------|-------|
| 1  | Discuss about the different types of windings in alternator                                                 | 3     |
| 2  | What are the causes of harmonics in the induced voltage of an alternator and how its effects are minimised? | 3     |
| 3  | Draw and explain the phasor diagram of a salient pole alternator supplying to a lagging power factor load   | 3     |
| 4  | Enumerate the requirement of proper parallel operation of alternators                                       | 3     |
| 5  | Discuss about any two starting methods of synchronous motors                                                | 3     |
| 6  | Derive expressions for starting torque in three phase induction motor                                       | 3     |
| 7  | Explain about working of star delta starter for three phase induction Motor                                 | 3     |
| 8  | Explain about V/f speed control method in three phase induction motors                                      | 3     |
| 9  | Compare the operation of induction Generator in grid connected mode and self excited mode                   | 3     |
| 10 | Explain about working of split phase induction motor                                                        | 3     |

**PART B**

(Answer one full question from each module, each question carries 14 marks)

**Module -1**

- |    |                                                                                                                                                |   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 11 | a) List the advantages of using stationary armature in alternator                                                                              | 7 |
|    | b) A 3 phase alternator, 50 Hz, 8 pole, has 180 conductors per phase, flux/pole is .054wb, find the induced line emf (assume $k_p = k_d = 1$ ) | 7 |
| 12 | a) What is the significance of winding factor in the induced voltage of induction motor? Derive an expression for the same                     | 7 |
|    | b) Explain different types of alternators based on rotor construction                                                                          | 7 |

**Module -2**

- |    |                                                                                                                                                                                                              |    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 13 | a) Find the full load regulation by MMF method of a 100 kVA, 2kV 3 phase 50 Hz star connected alternator at 0.8 power factor lag having the following test data, armature resistance = $0.2\Omega$ per phase | 14 |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|

**1100EET307122103**

|                   |     |      |      |      |      |      |
|-------------------|-----|------|------|------|------|------|
| If in A           | 10  | 20   | 25   | 30   | 40   | 50   |
| Vt (line voltage) | 800 | 1500 | 1760 | 2000 | 2350 | 2600 |
| Isc in A          | -   | 20   | 25   | 30   | -    | -    |

- 14 a) With circuit diagram explain about synchronisation of alternator using bright lamp method 7
- b) Explain slip test and calculation of  $X_d$  and  $X_q$  7

**Module -3**

- 15 a) A 75KW,400V,4 pole ,3 phase star connected synchronous motor has  $(0.04+j0.4)/\text{phase}$  ,compute for full load 0.8pf lead, armature current and gross mechanical power developed assuming an efficiency of 92.5% 7
- b) Draw & explain about the power- angle characteristics of synchronous motor 7
- 16 a) Discuss power stage diagram of induction motor 4
- b) The power input of a 500V,50Hz, 6 pole, 3 phase induction motor running at 975 rpm is 40KW,the stator losses is 1 KW and friction and windage losses are 2KW.find slip,rotor copper losses,BHP and efficiency 10

**Module -4**

- 17 a) Draw the circle diagram of a 20 HP,400V,50Hz,3 phase star connected induction motor from the following test data (line values) 14
- No load test : 400V 9A pf 0.2
- Blocked rotor test : 200V 50A pf 0.4
- from circle diagram find line current and power factor at full load and maximum power output
- 18 a) Explain about different braking methods of induction motor 7
- b) Explain how high starting torque and good operating efficiency is achieved in double cage induction motor 7

**Module -5**

- 19 a) Explain why single phase induction motor is not self starting 7
- b) Draw and discuss about speed - torque characteristics of induction generator 7
- 20 a) Explain about induction Generator 7
- b) Explain capacitor start capacitor run single phase induction motors 7